

THEO SIEMENS-RHODES

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EDUCATION

The University of British Columbia

BASc in Computer Engineering w/ High Distinction

Sep 2021 – May 2026

Vancouver BC, Canada

- GPA: 3.8/4.0, Dean's List 2021-22, 2022-23, 2023-24, 2024-25
- Relevant Coursework: Distributed Software Application: A+, Introduction to Cybersecurity: A+, Operating Systems: A, Machine Learning with Engineering Applications: A+, Digital Systems Design: A+, Computer Architecture: A

EXPERIENCE

SAP

Software Engineering Intern

Jan 2026 – Current

Vancouver BC, CA

- Migrated **Redis** caching layer from Jedis to **Lettuce** drivers with optimized multi-key operations, achieving **46% higher throughput** and **36% lower latency**; validated across **AWS, Azure, GCP, and Alicloud** with integration tests.
- Led development of **database observability** for **SAP HANA Cloud**, integrating metrics into **Dynatrace** dashboards; presented architecture at site-wide design review, closing a critical monitoring gap for **10+ microservices**.
- Drove **Java 17 to 25** runtime upgrade for the internal microservice framework used by **50+ services**, coordinating cross-team design and product reviews; enabled adoption of modern language features ahead of EOL deadline.
- Hardened service-to-service auth by implementing **JWT scope restrictions** to limit tokens to target services only, resolving a security vulnerability; added caller identification headers enabling **audit traceability** across microservice calls.

Hertzer Systems

Software Engineering Intern (Summers 2023, 2024, 2025, Fall 2025)

May 2023 – Dec 2025

Goshen IN, US

- Accelerated bulk CSV imports by over **500x** by offloading parsing to **C# .NET** backend task queue using MassTransit, removing per-row network bottlenecks and enabling fault-tolerant ingestion with support for partial retries.
- Implemented new **AngularJS** features, including audit dashboards and user license tracking; while also expanding support for user-defined scripting hooks, enabling greater customization and improving usability.
- Automated cross-platform installer builds for the companion local app by authoring an **Azure DevOps pipeline** to package platform-specific installers and publish artifacts to **Azure Blob Storage**, reducing manual deployment overhead.
- Optimized complex reporting queries in **PostgreSQL** by designing targeted indexes and partitioning large tables, reducing execution time to **300ms on datasets exceeding 1M rows**, which enabled real-time analytics and improved user experience.

UBC Launchpad

Technical Lead

Jan 2025 – Apr 2026

Vancouver BC, Canada

- Implemented **CI/CD pipelines in GitHub Actions** with automated rollouts, environment transitions, and secure config management, enabling zero-downtime deployments and reliable releases for full-stack **Dockerized** services.
- Reduced photo load times by **80%** by replacing inefficient JSON-encoded image payloads with **Express.js static file serving** and optimized asset delivery, significantly improving frontend responsiveness and overall user experience.
- Designed and implemented a full-stack **Typescript** audit logging pipeline where user actions triggered backend event records persisted in **MySQL**, providing comprehensive traceability of 10+ critical admin operations and strengthening accountability.

PROJECTS

RL Framework for AV1 Encoder Control | NetInt Technologies | C, C++, Python, Pybind11, OpenAI Gymnasium, SVT-AV1 2025

- Redesigned **SVT-AV1's** rate control using reinforcement learning, enabling dynamic superblock-level rate control adjustments and achieving up to 15% BD-Rate savings, delivering the same visual quality at significantly lower bitrates.
- Built a speed-optimized **C to Python** callback interface using **Pybind11**, allowing the encoder to make real-time model inference requests to a Python-based RL agent with minimal overhead and full multithreading support.
- Led policy architecture and episode threading, integrating **Stable-Baselines3** into a **OpenAI Gym-compatible** RL environment.

SKILLS & INTERESTS

Languages: Java, C#, Typescript/Javascript, Golang, Python, C++, C, SystemVerilog, HTML/CSS

Libraries/Frameworks: Spring Boot, Angular, React, Next.js, Prisma, Drizzle, .NET Core, Locust, JMeter, Jest, Playwright

Data/Infrastructure: Redis, PostgreSQL, SAP HANA Cloud, Solace, Dynatrace, Prometheus, Grafana Loki

Deployment: SAP BTP, AWS, Docker, Kubernetes, Cloud Foundry, Terraform, Traefik, Nginx, Github Actions

Developer Tools: Git, GitHub, Jira, Maven, JUnit, Bash, ModelSim, Quartus Prime

Interests: Rock Climbing, Jazz/Soul Music, Vinyl Listening, Personal Style/Thrifting